

LAB 8: WALLACE TREE ADDER

Duration: 2 hours

Marks: 10

Note:

- A manual on Verilog with details of available functions is also uploaded.
- The template, along with the autograder, is uploaded along with the question on Quanta.

Example: Add the following numbers: A <0101>, B<0001>, C<1000>, D<1100> and E<1101>

Step 1: Add D and E using carry look-ahead adder designed for adding two 4-bit numbers in previous lab.

D3	D2	D1	D0	1	1	0	0		
+	E3	E2	E1	E0	1	1	0	1	
Z4	Z3	Z2	Z1	Z0	1	1	0	0	1

Step2: Now to add A, B, and C we need 4 full adders as each number is of 4-bit. The output of this is X and Y.

	A3	A2	A1	A0	0	1	0	1	
+	B3	B2	B1	B0	0	0	0	1	
+	C3	C2	C1	C0	1	0	0	0	
Carry	X4	X3	X2	X1	0	0	0	1	
Sum	Y3	Y2	Y1	Y0	1	1	0	0	

X0 AND Y4 are set to zero to make them 5-bit numbers Thus X<00010> and Y<01100>

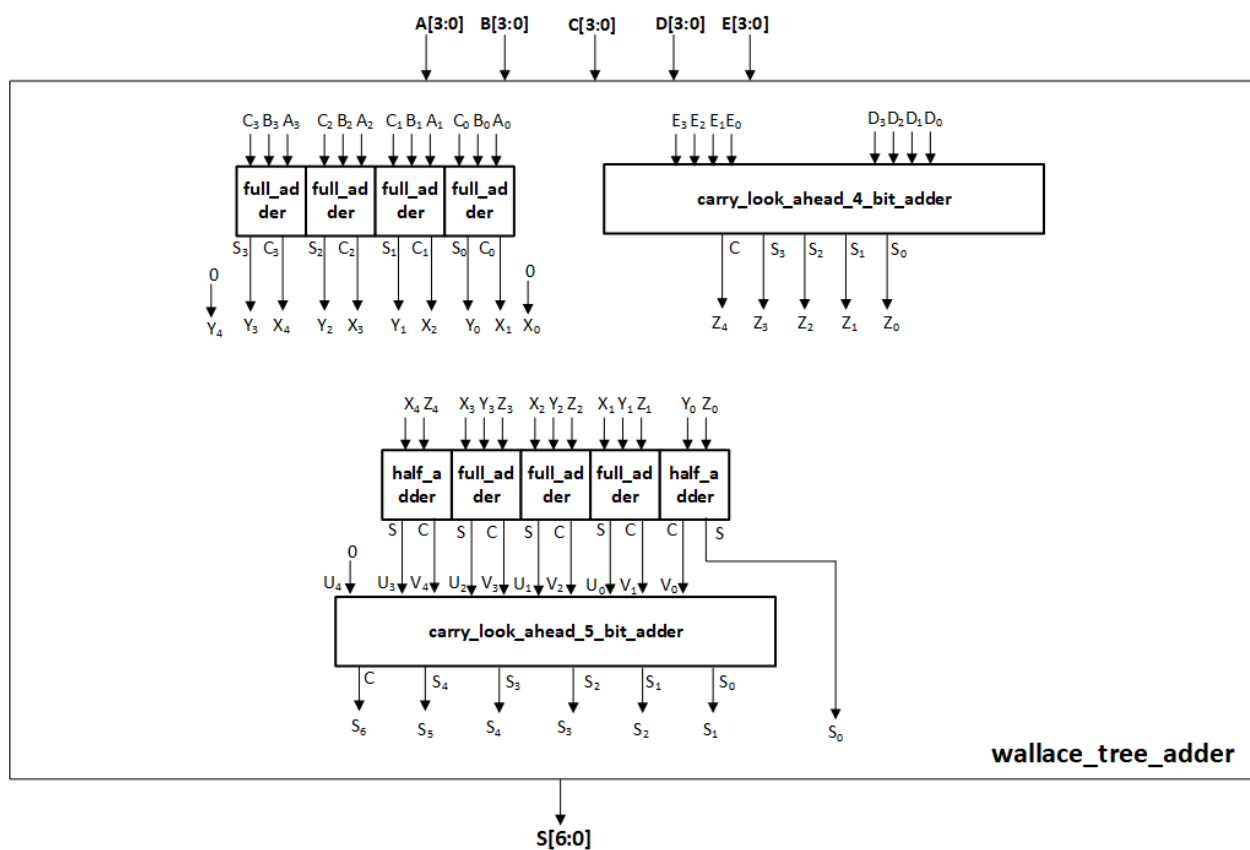
Step3: Now to add X, Y, and Z which are three 5-bit numbers.

	X4	X3	X2	X1	0	0	0	1	0			
+	0	Y3	Y2	Y1	Y0	0	1	1	0	0		
+	Z4	Z3	Z2	Z1	Z0	1	1	0	0	1		
CARRY	V4	V3	V2	V1	V0	0	1	0	0	0		
SUM	U4	U3	U2	U1	U0	S0	0	1	0	1	1	1

Step 4: Add U and V using 5-bit carry-lookahead adder

$$\begin{array}{r}
 \begin{array}{cccccc}
 & U4 & U3 & U2 & U1 & U0 \\
 + & V4 & V3 & V2 & V1 & V0 \\
 \hline
 S6 & S5 & S4 & S3 & S2 & S1 & S0
 \end{array}
 &
 \begin{array}{r}
 \begin{array}{ccccc}
 & 0 & 1 & 0 & 1 & 1 \\
 & 0 & 1 & 0 & 0 & 0 \\
 \hline
 0 & 1 & 0 & 0 & 1 & 1 & 1
 \end{array}
 \end{array}
 \end{array}$$

Circuit for Wallace Tree Adder



Instructions-

- 1) After downloading template zip rename with your BITS ID and need to submit the same folder by compressing it to zip again. ONLY use ZIP format.
E.g. <BITS_ID>.ZIP
- 2) Run the testcases using **python3 run_verilog_tests.py** or **python run_verilog_tests.py**
- 3) Mark distribution is as follows:
half_adder.v - 1 mark

full_adder.v - **1 mark**

Carry_look_ahead_4_bit_adder.v - **2.5 mark**

Carry_look_ahead_5_bit_adder.v - **2.5 mark**

Wallace_tree_adder.v - **3 mark**